Baltimore Orioles: An Analysis of the First-Inning Home Advantage

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1. Introduction

In many sports, the home team often has an advantage, whether due to crowd support, familiarity with the playing field, or other factors. Specific to Major League Baseball (MLB), home teams win approximately 54% of their games. However, in the MLB's storied history, this advantage is most pronounced in the first inning, in which the home team outperforms the visiting team at a significantly higher rate relative to other innings. This paper investigates the phenomenon of the home-field advantage in the first inning of MLB games.

Existing literature by David W. Smith published in 2014 and 2015 indicates that though the home team has a pronounced advantage in all innings, the advantage is most pronounced in the first inning. Additionally, he found that travel does not explain the difference in scoring, and the best explanation for the advantage seems to be greater length of the top of the first, which is correlated with scoring in the bottom of the first.

We first extend this previous research using data from the past decade to confirm whether or not home teams still have an advantage and that such an advantage is most substantial in the first inning. Building upon this result, we explore potential causal factors of the effect with an emphasis on travel-related factors. We find that although travel has a minimal effect, it does not fully explain the observed advantage. Finally, we consider first-inning scoring as a prediction problem, utilizing regression and machine learning models to predict whether or not either team will score in the first inning with NRFI (No Run First Inning).

2. Methods

To obtain our main objective in determining if a first inning advantage still exists today and if this advantage exists largely in the first relative to other innings, we replicated David Smith's RetroSheet graphs with modern data dating from 2013-2024. To extend our analysis, we also explored possible causal factors and then posed this as a prediction problem.

2.1 Modern Data Graph Replication

We first downloaded play-by-play data from 1910 to 2024 and box score data from 1898 to 2024 from RetroSheet, which provided us with general information (e.g. date, final score, inning-by-inning scoring) and more specific data (e.g. field conditions, pitcher information). It is important to note that we used data from years prior to our main study period of 2013-2024 so we could reproduce Smith's 1909-2013 results and verify that our code and models matched his before applying them to modern seasons. Since RetroSheet data isn't neatly formatted into analysis-ready csv files, we ran our scraper on event (play-by-play) files to output four files for each year: "games.csv," "plays.csv," "roster.csv," and "rosters.csv." "games" summarizes each game in one row, providing information like game ID, home and visiting team, winning and losing pitchers, and start time. “plays” goes more in-depth, including the raw event description of each play, with information like inning and batter count. “roster” and “rosters” both contain player information, with “roster” containing information on each player per game, and “rosters” having information on each player per season.

Then, with these neatly arranged files for each season, we can replicate the figures created by David W. Smith. We began by processing each game’s play-by-play data to calculate the runs scored by home and visiting teams in each inning. For each year, it sums runs across games, computes averages for home and away teams per inning, and calculates the probability the home team scores, the probability the visiting team scores, and the first-inning run differential. These results are written to a csv file, which is formatted as one row per inning per year, with columns for number of games, average runs for home and visiting teams, probability of scoring, and run differential. Using the summary csv file, we generated Figure 1 by calculating the average runs scored by home and visiting teams across all innings for the selected years. Figure 2 shows the average runs for home and visiting teams separately, and we generate Figure 3 by plotting the differential between these averages. Figure 4 visualizes the run differential for each year, directly using the run differential column for each year. Finally, Figure 5 aggregates the first-inning run differential by decade from 1910 to 2024, computing weighted averages using the number of games per year to ensure that years with more games contribute proportionally to the overall trend.

2.2 Possible Driving Factors

Travel.

After confirming that the first inning home-field advantage still exists in the modern day, we moved on to considering travel as a potential cause of the advantage. We first load game-level data for each year from 2014 to 2024, including the game ID, date, home team, visiting team, and first-inning runs for both teams, while ignoring missing files or incomplete data. Using this information, we calculated the distance traveled by both the home and visiting teams since their previous games with a Haversine function.

Specifically, for each team, the script retrieves the latitude and longitude of the previous stadium and the current stadium, converts the longitudinal and latitudinal differences to radians, computes the central angle between the points using the Haversine formula, and multiplies by the Earth’s approximate radius to obtain the distance in kilometers. Stadium coordinates were obtained from publicly available sources and stored in a Python dictionary mapping each team ID to its latitude and longitude.

These travel distances were then used to flag the first game of a series: any game in which either the home or visiting team traveled more than a minimal threshold (0.1 km). All of this processed information was compiled into a single pandas DataFrame and exported to a CSV. This structured dataset served as the input for subsequent negative binomial regression analysis to evaluate the relationship between travel and first-inning home-field advantage.

From here, we fit a negative binomial regression model to the first-inning run counts, both with and without travel as a covariate. We set the response variable as the number of runs scored in the first inning, a non-negative discrete variable. However, a standard Poisson model would not be appropriate to calculate expected runs, as runs in the first inning are overdispersed: the data have a large mass at 0 and a variance larger than its mean. The negative binomial model directly addresses this issue with its additional dispersion parameter, which allows for the variance to exceed the mean like is the case for first inning runs. Our model works in the following way:

For each game, we have two observations: one for the home team and one for the visiting team. For one observation i, we let Y_i denote the number of first-inning runs and X_i denote the vector of covariates, which includes whether the team is at home, and for when the model is run on travel data, the distance traveled. Then, we have

$$Y_i | X_i \sim \text{NegBinom}(\mu_i, \alpha)$$

where we have

$$E[Y_i | X_i] = \mu_i \text{ and } Var[Y_i | X_i] = \mu_i + \alpha\mu_i^2$$

Here, $\alpha > 0$ is our overdispersion parameter. We then model the mean with a log link (to ensure the mean is positive) in the two following scenarios:

Without travel:
$\log(\mu_i) = \beta_0 + \beta_1(Home_i)$

With travel:
$\log(\mu_i) = \beta_0 + \beta_1(Home_i) + \beta_2(Travel_i)$

Here, $Home_i$ takes on 1 if it's a home game and 0 if it's an away game. $Travel_i$ is a standardized measure of the distance a team traveled to reach the stadium for the game, particularly:

$$Travel_i = \frac{distance\ traveled\ by\ team - mean\ travel\ distance}{standard\ deviation\ of\ travel\ distance}$$

The model estimates the values for α and β based on maximum likelihood, which finds the values that make the observed data most probable under the negative binomial model. This allows the model to produce more accurate standard errors and predictions than a standard Poisson model. Thankfully, python's "model.fit" automatically performs this maximum likelihood calculation for us. The values of each β indicate to what extent each factor has on scoring in the first inning. Thus, the higher the magnitude of each β, the greater the effect it has on scoring.

For each game, we create two observations: one for the home team and one for the away team, allowing the home indicator and travel covariate to be assigned correctly for each observation. Then we fit the negative binomial model to these observations to estimate expected first-inning runs for home and away teams, both with and without travel as a covariate, and compute the home-field advantage as the difference between these expected values, outputting the expected home and away runs, the difference, and β values to a csv.

Time-Zone Effects.

To further our analysis with travel, which resulted in having not as much of a significant effect, we turned to time-zone displacement and examined the relationship with home team advantage. We created a dataset that includes 2013 through 2024 seasons and created a time-zone variable that accounts for both the magnitude and direction of circadian disruption that the visiting team may experience. We obtained raw game metadata from season-level game log files which includes game identifiers, game dates, home and visiting team identifiers, and venue information. We standardized these files across seasons by keeping the columns that were necessary for chronological ordering and matchup by parsing all dates into a consistent datetime format. We then merged the game records with a separate results file which contained finalized run totals for both home and visiting teams. We then constructed a binary outcome variable which indicates a home-team win, where “1” represents the home team’s final score exceeding the visiting team’s final score, and “0” otherwise. We excluded games with missing, invalid, or zero-sum scores to ensure that our data represented complete and valid outcomes.

To observe and isolate the effects of time-zone adjustments, we first identified series-opening games since these games are likely to reflect minimal travel effects. We grouped games by home team, visiting team, and venue, and marked a game as a series opener if it was the first occurrence of a given matchup or if there was a gap of one day. For each remaining game, we quantified the visiting team’s travel by distance traveled and time-zone displacement. We use the haversine to calculate travel distance based on the latitude and longitude coordinates assigned to each team’s home ballpark. For each visiting team, we ordered games chronologically, and the distance traveled prior to a game was calculated as the distance between the location of the team’s previous game and the location of the current game.

We assigned each team a fixed time-zone offset based on the location of its home ballpark, relative to Coordinated Universal Time, to calculate time-zone displacement. For every game, we computed a signed time-zone difference as the home team’s time-zone offset minus the visiting team’s offset. Positive values indicate west-to-east travel while negative values indicate east-to-wet travel. 0 represents games played between teams in the same time zone and serves as our baseline. We restricted our dataset to series-opening games that had significant travel, to which we defined as a distance of at least 1,000 kilometers by the visiting team since its previous game or an absolute time-zone difference of at least one hour between the home and visiting teams as small travel events are highly unlikely to produce such effects. We saved this filtered dataset and leveraged it to our analysis.

We estimated the correlation between time-zone displacement and win outcomes by grouping games by their signed time-zone difference and calculated the mean of the home-win indicated for each group to compute the home-team win probabilities. Since our outcome variable is binary, the grouped means correspond to the estimate of the probability of a home-team win condition given direction of time-zone displacement. We visualized these estimates by plotting home-team winning percentage against time-zone difference.

Bullpen “Cool Down” Effect.

We also explored the “cool down” effect in which when the top of the 1st inning takes a greater length of time, the visiting starting pitcher waits longer before pitching the bottom of the 1st, which could potentially lead to cooling down and losing effectiveness.

We constructed a game-level dataset using pitch-timing and outcome information from MLB games spanning the 2015 season to the most recent available season. For each game, we measured the elapsed duration of the top of the first inning in which we calculated the difference between the first and final pitch of the top half inning, using pitch-level timing information aggregated to the game level. We recorded this relapsed time variable in seconds to capture the length of time in which the visiting starting pitcher is inactive between completing pregame-warm-up and taking the mound for the bottom of the first inning.

We observed whether or not the home team scored at least one run in the bottom of the first inning from the official run totals and encoded this as a binary indicator where “1” indicates the home team scored one or more runs in the bottom of the first inning and “0” otherwise. This allowed us to directly correspond to the visiting pitcher’s failure to “escape” the first inning without allowing a run. We converted this outcome into a complementary “visitor escape rate” with 1- P(Home Scoring), representing the probability the visiting starting pitcher allowed 0 runs in the bottom of the first inning.

We grouped games into quintiles based on the distribution of elapsed time, using equal-frequency cuts in which each bind contained an equal number of games so that we can ensure a comparable sample size across bins. The first quintile represents games with the shortest top-of-the-first elapsed times and the fifth quintile represents games with the longest elapsed time. For each quintile, we computed three statistics: the total number of games, the proportion of games where the home team scored in the bottom of the first inning, and the mean elapsed time in seconds. We wrote these summaries to an intermediate aggregated file which we used as a direct input to our visualization.

We plotted the visiting pitcher escape rate as a percentage against the elapsed-time quintile. Each plotted point corresponds to the mean escape probability within a quintile, and we annotated each point with the average elapsed time measured in seconds for that bin. Because the outcome is binary at the game level, the means correspond to empirical probabilities conditional on elapsed time.

Umpire Bias.

We scraped data from StatCast pitch-by-pitch data search via Baseball Savant. We approached calculating umpire bias by focusing on the left and right side of the strike zone because umpire subjectivity is relatively strongest there. From their offset stance, vertical calls are relatively consistent, but horizontal edges are visually ambiguous and more affected by framing and crowd influence. MLB’s Statcast and Baseball Savant datasets also show that over 75% of “borderline” missed calls occur on the horizontal edges (not top/bottom), and the top and bottom of the zone tend to have smaller error margins, partly because pitch-tracking calibration uses the batter’s knees and chest as clear reference points. We approached this by calculating the the over-expected call strike in which we calculated the strike zone geometry as so:

$$\text{Vertical Midpoint for each Pitch: } Z_{center}=(sz_{top}+sz_{bottom})/2$$
$$\text{Vertical Location Relative to Zone: } Z_{rel}=plate_z-Z_{center}$$

We then trained a logistic regression model on called pitches only, excluding swings, to estimate the baseline probability that a pitch is called a strike given its location and context:

$$\hat{p}_i=P(strike_i \mid X_i)= \frac{1}{1+e^{-(\beta 0+\beta 1 platex+\beta 2 Zrel+\beta 3 balls+\beta 4 strikes+\cdots)}}$$

Then, we computed the over-expected called strike. For each pitch i, we let y_i = 1if called strike and 0 if called ball.

$$OCS_i = y_i - \hat{p}_i$$

If OCS > 0, there were more called strikes than expected. If OCS < 0, there were less strikes called than expected. We applied fringe zone filtering as human judgement is required here to isolate borderline pitches where umpire discretion matters to the value of:

$$0.3 \le \hat{p}_i \le 0.7$$

2.3 The Prediction Problem

Before creating our models to predict first-inning scoring, we first took game-level data, including travel distance, first-inning runs, game ID, and home and visiting team from our previous model's travel csv. Then, from Baseball Reference, we downloaded pitcher data, including ERA, from the years 2014 to 2024. Then, matching pitcher names from our "rosters.csv" file from each year, and creating a dictionary mapping certain common errors (e.g. dropping accents and changing nicknames), we outputted the ERA for each pitcher into our game level data. Then, we created a script to calculate the average runs in the first inning for the past ten games. If a team didn't play ten games yet, it averages all its past few games, and any remaining missing values are just filled with its observed first-inning runs so no row is unfilled. We also inputted home and away on-base percentage uploaded from TeamRankings.com. An important consideration is that this information is for the season in totality, meaning that it is not the current on-base percentage of the game itself. As a result, in order to predict future games, we will need to fit the model instead with current OBP. We merge each year's csv into one large csv with game ID, home and visiting team, home and visiting first inning runs, ERAs for both pitchers, home and visiting travel, home and visiting average runs in the first inning for the past ten games, and home and away OBP.

Logistic Regression.

Given this data, our logistic regression model predicts the probability that a run is scored in the first inning through considering YRFI (yes run in first inning) as a binary outcome. The response variable is defined as one when at least one run is scored in the first inning and zero otherwise. Our predictors are all of the data in the merged csv, with our training set being all years up until 2024 and our testing set being 2024 data. Continuous variables are rescaled to a common scale, and missing numerical values are filled in using median values from the data. For categorical variables, any missing entries are replaced with the most common category observed, after which these variables are encoded for use in the model.

We apply year-decay sample weights during model fitting, assigning greater weight to more recent seasons. This will allow for more recent data to have a greater impact on the model, allowing for us to control for rule changes and other season-by-season changes. The data are split in a time-aware manner, with the most recent season reserved for testing when possible. Model performance is assessed using several complementary metrics, including ROC-AUC, log loss, F1 score, and precision–recall AUC, which together capture both overall discrimination and the quality of the predicted probabilities. ROC-AUC measures how well the model separates games where a run occurs vs. where a run does not. Ideally we want a value closer to 1, and a value of 0.5 means that the model is essentially random guessing. Log loss measures the quality of probabilities, penalizing confidently incorrect predictions more than uncertain ones. We want a lower value for this, but values around 0.55 to 0.65 are reasonable and informative. The F1 score measures how well the model balances correctly predicting first-inning runs while avoiding too many false positives. We want a score in the 0.45 to 0.55 range, which is relatively strong. Precision–recall AUC considers how well the model can identify first-inning run games by balancing how often its positive predictions are correct against how many true scoring games it successfully finds. A higher score for this is better, and 0.50 is around random scoring.

To make the results easier to interpret, predicted probabilities are then converted into binary predictions using a fixed classification threshold of 0.50, allowing us to directly compare predicted first-inning scoring outcomes with observed game results.

XGBoost.

Our XGBoost model also predicts the probability of YRFI as a binary outcome (either YRFI or NRFI), but instead of assuming a mostly linear relationship between predictors and scoring, it learns flexible, non-linear patterns using an ensemble of decision trees. We use the same merged dataset and preprocessing pipeline as the logistic regression, with our training set being all years up until 2024 and our testing set being 2024 data. XGBoost then repeatedly creates decision trees that split on these features (for example, it changes outcomes based on if travel is greater than a certain threshold), allowing the model to capture certain interactions in combination, like "travel matters more when the visiting team has less rest."

The model is trained as a sequence of trees, where each new tree is added to correct the mistakes of the trees built so far, a machine learning algorithm. In our implementation we set the combined output of all trees to be transformed into a probability between 0 and 1. We control model complexity by limiting each tree's depth to 4, meaning each tree can only make at-most four sequential splits, making each tree simple and reducing hyperspecific models that overfit. We also update the model gradually with a small learning rate of 0.05 over 600 boosting rounds. This will allow the model to slowly improve without making large, unstable updates. We use random subsampling of rows and features when building each tree in order to improve generalization and further reduce overfitting, so that the model will not rely too heavily on a single subset of games or predictors. We include L2 regularization, which prevents too complex trees from forming, allowing for further stability. As with logistic regression, we apply year-decay sample weights during fitting so that recent seasons affect the splits more strongly, and we output ROC-AUC, log loss, F1, and precision–recall AUC. Finally, predicted probabilities are converted into binary predictions using the same fixed 0.50 threshold so we can directly compare the model's "YRFI vs. NRFI" calls to the observed outcomes.

Test Cases.

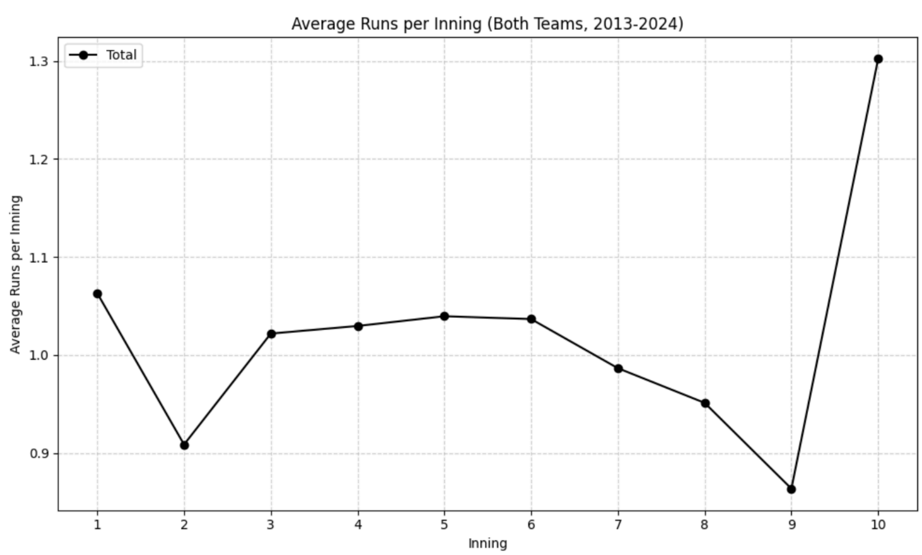
For our test cases, we went to the Internet Archive to get raw betting data from MGM sportsbook, which gives NRFI odds for all games on a specific date. Then, since there is not really any great way to get an automated pipeline from Internet Archive, we created a game_ids csv file with the game IDs of all the day's games, the given betting odds, and MGM's picks. We then loaded our pre-trained models from the .joblib pipeline files, applying them directly to the corresponding game-level features to generate first-inning scoring probabilities. Then, we converted these probabilities in a similar way, with probabilities >0.5 being counted as YRFI and <0.5 as NRFI. Then, with the odds and a $10 flat bet (with the assumption that YRFI odds are the same as NRFI but multiplied by -1), we calculate the expected profit, outputting that as a csv.

3. Results

We have successfully confirmed our hypothesis by replicating David Smith's graphs with modern data, picking up where he had left off. From the years 2013-2024, we are able to see several clear patterns that affirms our hypothesis: there still exists a first inning advantage today and that advantage exists largely in the first inning relative to other innings. We included several figures below that strongly support this claim. In this section, we will also dive in-depth into the possible factors that drive this advantage, in which we rule out travel and hypothesize that the first inning advantage results from a multitude of factors.

3.1 Modern Data Replication

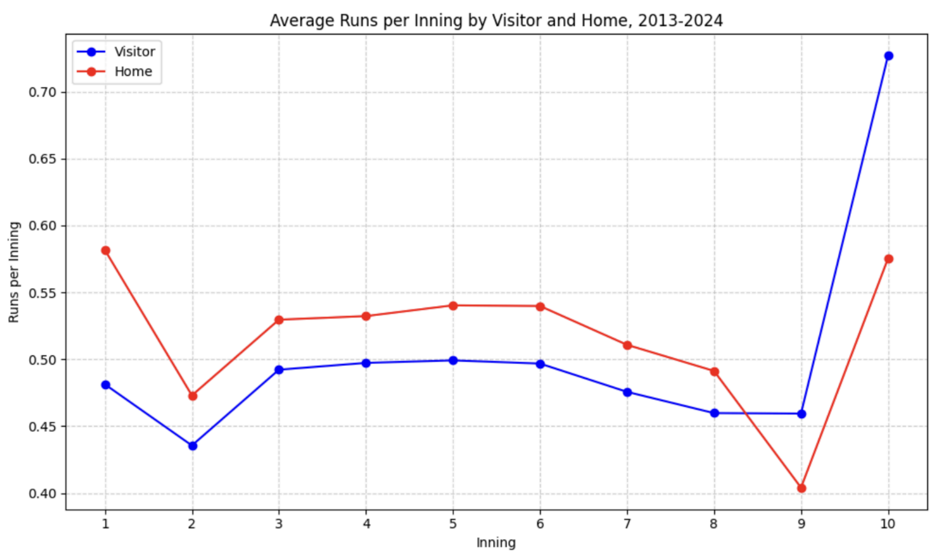
Figure 1.



Similar to David Smith's findings for 1909-2013, we found that from the years 2013-2024, the first and second innings contrast quite visibly, in which the first inning has the highest average runs per inning and the average runs drops for the second inning. We see a relatively constant pattern from the third inning and onwards, in which we see a dropoff at the ninth inning. This can be explained by the fact that if the home team leads at the bottom of the ninth, they do not bat and thus win the game without playing the rest of the inning, lowering the average runs in the inning. However, we see a major spike in the tenth inning, differing from Smith's findings, which can be explained by the implementation of the "ghost runner" rule in 2020. As a result, a runner is placed on second base during extra innings (from the tenth and on), greatly increasing the odds that a run is scored.

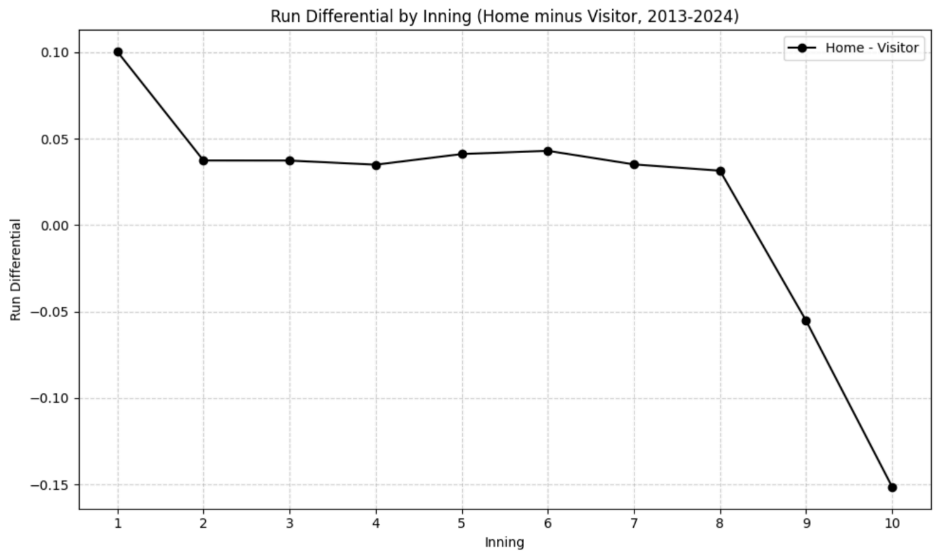
We also divided the data into home and visiting teams which helps visualize the differences in their scoring rates, as displayed below in Figure 2. The home team is shown in red and the visiting team is shown in blue.

Figure 2.



The lines are generally parallel to one another until approximately the ninth inning, which can be similarly explained by the fact that home teams don't bat when leading at the bottom of the ninth and the new ghost runner rule's implementation in 2020. We can further analyze this behavior by observing the difference in runs scored between the home and visiting team in Figure 3 below.

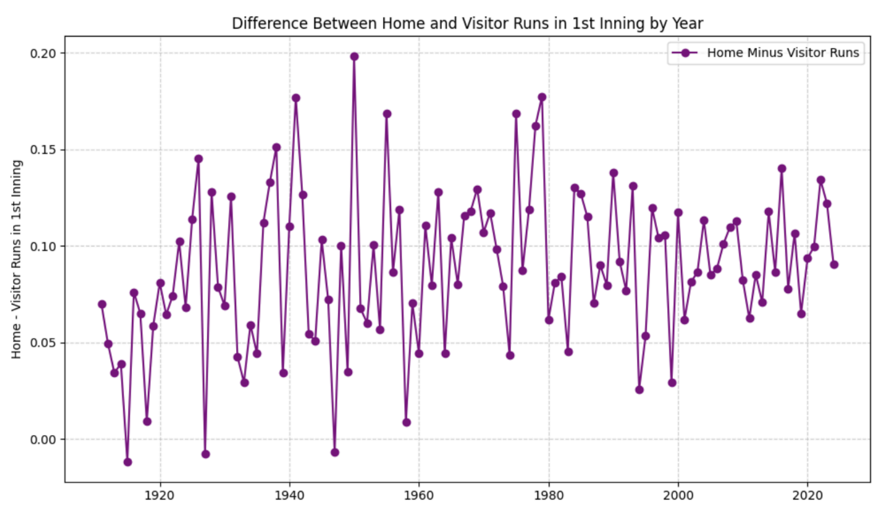
Figure 3.



To reiterate David Smith's point, Figure 2 appears deceiving due to the parallel trend between the home and visiting teams, and we can now clearly see that the home team scores significantly more runs in the first inning than the visiting team. The differential in this graph exemplifies the advantage the home team holds across innings – the advantage being most significant in the first inning.

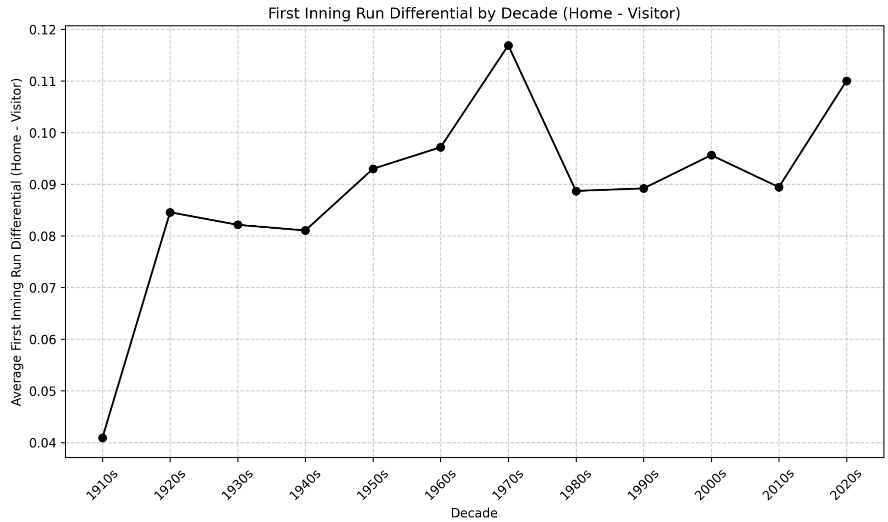
Next, we determined if the first inning advantage is consistent through all years. We will now examine and extend the variability from where Smith had left off in Figures 4 and 5.

Figure 4.



Though there is a significant amount of annual variation, as expected, we observe that in 113 of the 116 seasons, the home team scores more than the visiting team in the first inning. We then now grouped the years into decades to see a clearer visualization and decrease the noise.

Figure 5.



By grouping the yearly data into decades, we now see a clear pattern. The drop between 1910-1920 can be explained by the "dead ball era" in which teams averaged very few runs. The jump after the 1920s can be explained by the "lively ball era" in which a series of rule changes helped make the ball more "lively." We see relative consistency throughout the decades from thereon, with a clear spike in the 1970s and a climbing peak in the 2020s. However, we do not have a definite answer today as to why we observe these spikes, though for the 2020s, we can cautiously hypothesize that the ghost-runner implementation may have quite a bit of an impact.

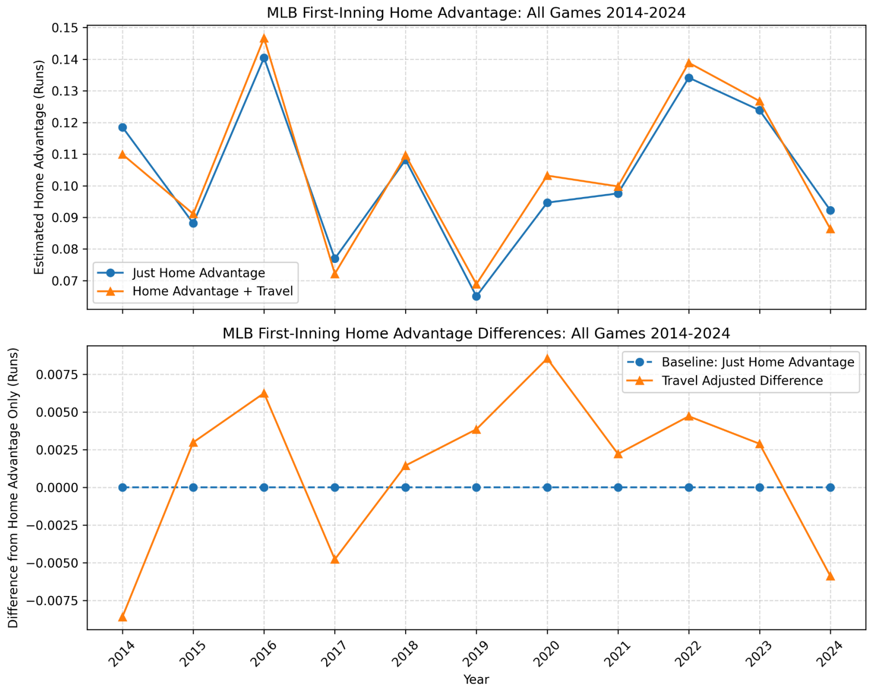
It is evident that the first inning home advantage still exists in modern years and perhaps understanding what may be causing this advantage would be beneficial. We pivoted to examining travel as a potential driving factor, to which David Smith had disproven as having only a minimal effect.

3.2 Possible Driving Factors

After recreating Smith's work on travel's effect on the overall advantage home teams have, we decided to look into its effect on just the first inning, before looking at other possible factors.

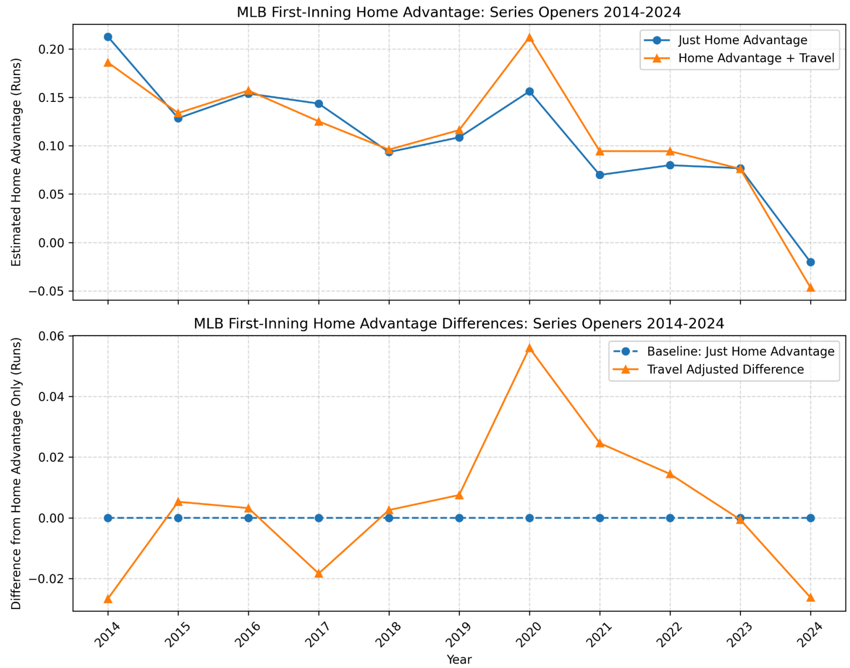
Travel

Figure 6.



Year-by-year, we can see that accounting for travel, each year's estimated home advantage from our negative binomial model seems to take similar values as when travel is not accounted for. When we take the estimated home advantage as a baseline, the variations from each year seem randomly distributed, and only affect the estimated home advantage by a very small amount. In fact, estimated home advantage with travel is always contained within a range of 0.0055 points from the estimated home advantage. If travel had a significant effect on first inning home advantage, we would see it be consistently lower than the baseline, which is not the case here. In fact, the outputted β-value for travel for all years has a magnitude of less than 0.005. Thus, travel does not seem to have a significant effect on home-field advantage in the first inning. However, since games are played in a series format, teams stay in the same location for three to four games, so the majority of games involve no travel at all.

Figure 7.



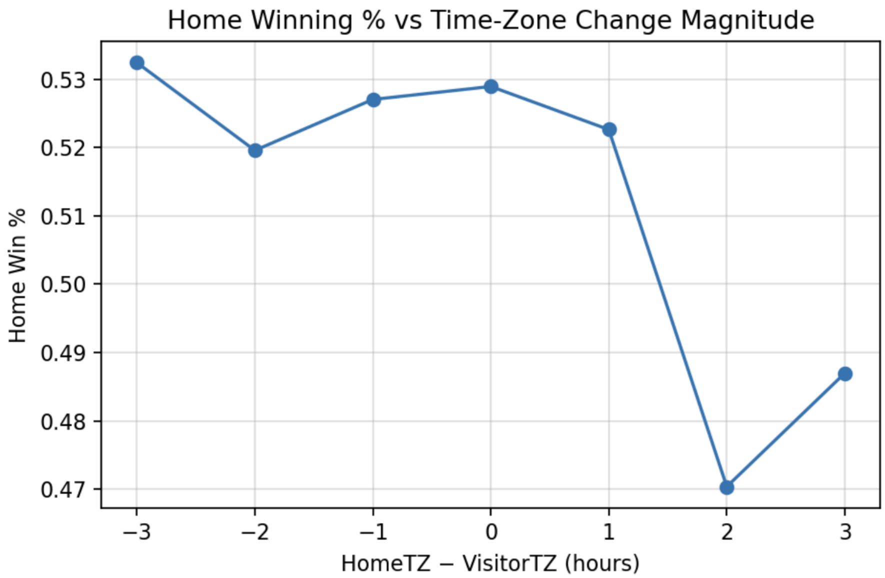
When we filter the games to just series openers, we see that the estimated home advantage is more sporadic, taking values from an advantage of 0.2 runs to even a lack of a home advantage in 2024. When our model is adjusted with the mean travel each year, we see some larger but still modest shifts in the estimated home advantage in the first. Some years it amplifies it and some years it lessens it, but it does not systematically explain the advantage. Thus, travel might have some effect on home-field advantage in the first inning, but does not fully explain the advantage. Some of the noise may also be explained by a somewhat smaller sample size: of the over 2,000 regular season games, only around 700 are series openers.

After ruling out travel as a driving factor, we examined potential other factors such as the "cool down" effect in the bullpen, umpire bias on strike calls, time-zone effects, etc.

Time-Zone Effects.

Though we ruled out distance traveled as a significant factor, we believed it would be interesting to see if traveling east-to-west or west-to-east would have a more pronounced impact due to time-zone changes and adjustments players would potentially have to account for.

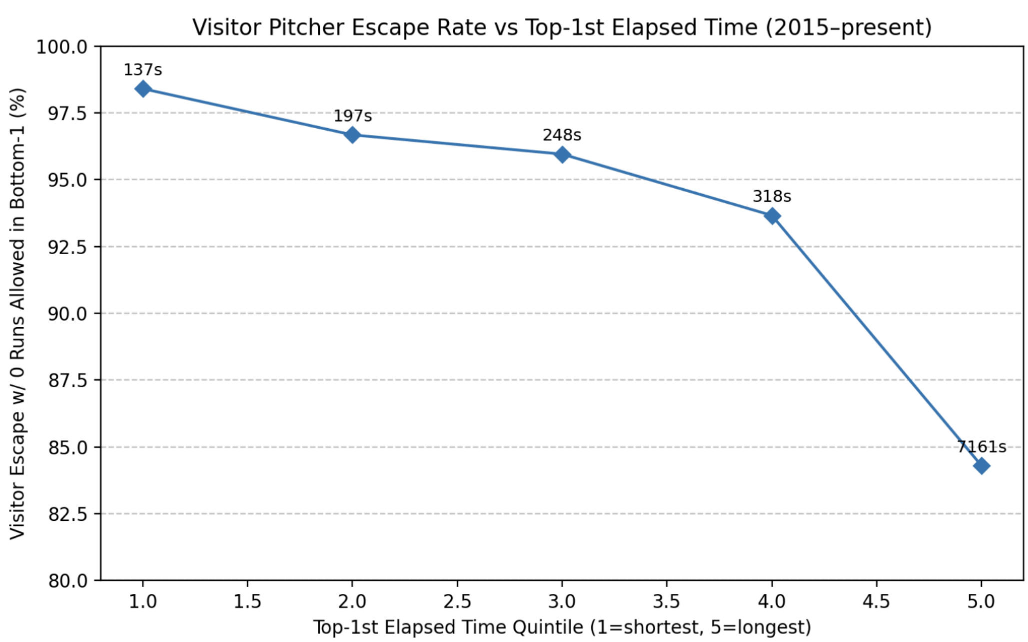
Figure 8.



While time-zone differences were examined as a potential mechanism, we find limited and inconsistent evidence that eastward travel alone produces systematic first-inning disadvantages. Any observed effects are context-dependent and do not persist across innings or translate reliably to full-game outcomes.

Bullpen "Cool Down" Effect.

Figure 9.

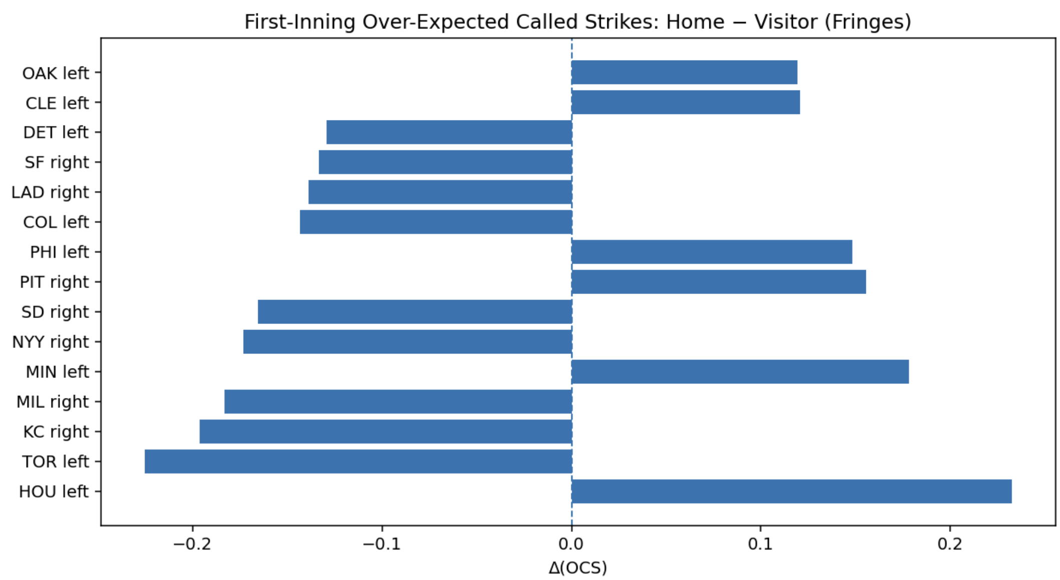


We find a strong monotonic association between the length of the top of the first inning and visiting pitcher performance in the bottom of the first, with longer top-of-first innings corresponding to lower escape rates. As so, we see that greater length of the top of the first inning is strongly associated with increased scoring in the bottom of the first.

Umpire Bias.

We decided to focus on the left and right edges of the strike zone because that's where umpire subjectivity actually shows up. From their offset stance, vertical calls are relatively consistent, but horizontal edges are visually ambiguous and more affected by framing and crowd influence. MLB's Statcast and Baseball Savant datasets also show how over 75% of "borderline" missed calls occur on the horizontal edges (not top/bottom), and the top and bottom of the zone tend to have smaller error margins because pitch-tracking calibration uses the batter's knees and chest as clear reference points.

Figure 10.



HOU left (23% more left edge strikes for Houston's home pitchers than model predicted - strong home bias)
TOR left (22% fewer left edge strikes for Toronto's home pitchers than expected - visitors got more generous left edge calls)
NYY right (17% less right edge calls which slightly favors visiting pitchers instead)

We find evidence of asymmetric first-inning strike calls on borderline pitches between home and visiting pitchers. However, the direction and magnitude of these differences vary substantially across teams and handedness and do not indicate a uniform home-field advantage.

3.3 Predictive Models

We pivoted to see how well we can anticipate whether the home team scores in the first inning of a given game, and we will examine the results of our models and compare them to the actual scores.

Our Logistic Regression and XGBoost models have a general accuracy rate of 59.9% and 70.6% respectively when calculating home NRFI. Both models have ROC-AUC values well above 0.5, indicating that they can reliably distinguish between games with and without first-inning runs. XGBoost achieves higher ROC-AUC, lower log loss, and a stronger precision–recall AUC, suggesting that it has better overall discrimination and probability calibration, while logistic regression attains a slightly higher F1 score, reflecting a better balance between precision and recall at the fixed 0.50 threshold. Overall, both models show meaningful predictive ability.

Test Case: September 14, 2024.

For all the games on this date, logistic regression had an accuracy rate of 60% and XGBoost of 46.67%.

Now, if we extended this to sports betting:

Payout Results 1.

game_id	nrfi_odds	bet	logreg_pick	logreg_payout	xg_boost_pick	xg_boost_payout	mgm_pick	mgm_payout	result
PIT202409140	100	10.0	yes	-10.0	yes	-10.0	no	10.0	no
SFN202409140	-105	10.0	no	10.5	yes	-10.0	no	10.5	no
CLE 202409140.00	105	10.0	yes	10.5	yes	10.5	no	-10.0	yes
SEA202409140	130	10.0	yes	13.0	yes	13.0	no	-10.0	yes
TOR202409140	105	10.0	yes	10.5	yes	10.5	no	-10.0	yes
COL202409140	-150	10.0	yes	-10.0	yes	-10.0	no	15.0	no
DET202409140	100	10.0	no	-10.0	no	-10.0	no	-10.0	yes
ATL202409140	110	10.0	yes	-10.0	yes	-10.0	no	9.09	no
ANA202409140	-105	10.0	no	10.5	no	10.5	no	10.5	no
MIN202409140	-105	10.0	yes	-10.0	yes	-10.0	no	10.5	no
NYA202409140	-105	10.0	yes	9.52	yes	9.52	yes	9.52	yes
WAS202409140	-115	10.0	no	-10.0	yes	8.7	no	-10.0	yes
ARI202409140	-105	10.0	no	10.5	yes	-10.0	yes	-10.0	no
PHI202409140	-105	10.0	no	10.5	yes	-10.0	no	10.5	no
CHA202409140	-105	10.0	yes	9.52	yes	9.52	no	-10.0	yes

Total Profit ($10 flat bets)

LogReg: $35.04

XGBoost: $-7.76

MGM: $15.61

We see that our logistic regression model outperforms MGM's payout of $15.61, but XGBoost does not make a profit at all. Generally, it is the case that our XGBoost model is more accurate than logistic regression, but this is a rare exception where logistic regression works better.

Though our data was very limited, we can see that our models are comparable to those by professional sports books. Our models are relatively accurate, though some fine tuning would immensely improve our results.

4. Conclusion

Our conclusion aligns with David Smith's prior research: there exists a first inning home advantage. In modern data, it appears that there exists a larger advantage specifically in the first inning compared to other innings more so today than there was in the past, probably due to the ghost runner implementation in 2020 improving overall number of runs scored in games. Similar to Smith's conclusion, we can also rule out that the distance a team travels has little to no effect on this advantage and does not explain the scoring difference. However, there is no singular cause that has yet to be verified to be the driving factor that explains this difference, though we hypothesize that the advantage is a result of a multitude of factors in conjunction, namely, rest between games, travel distance, and rest for pitchers. Our research suggests that the first inning advantage is a clear and consistent behavior that is present throughout all MLB data, though it now extends to and poses the question: where exactly is the run scoring advantage stemming from?